

Supplementary Materials

Mechanism-Dependent Validity of *Cis*-pQTL Mendelian Randomization
for Drug-Target Prioritization

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Contents

1 Protocol Deviation Log

The protocol evolved across six publicly versioned iterations, each SHA-256 hashed before execution. All amendments were made before outcome adjudication and logged with timestamps.

Table 1: Table S1: Pre-registration protocol versions.

Version	Frozen	Key changes
V1	2026-07-04 04:57Z	Initial 3-class taxonomy; 3 prospective predictions
V2	2026-07-04 07:15Z	6-class taxonomy; binary endpoint; 8 diseases
V3	—	EpiGraphDB <i>cis</i> -pQTL catalog; <i>cis</i> -only single-SNP Wald ratios
V3.1	—	Peer-review fixes; $n = 4$ (underpowered)
V3.2	—	16 diseases; $n = 30$ (BA = 0.589)
V3.3	—	33 diseases; $n = 50$ (BA = 0.580)
V3.4	2026-07-04 23:51Z	UKB-PPP instruments; 24 diseases; $n = 195$ evaluable (138 analyzed)

What did not change across versions. The classifier ($p < 0.05 \rightarrow \text{CAUSAL}$), the instrument type (*cis*-pQTL, single-SNP Wald ratio), the outcome adjudication procedure (FDA/EMA and ClinicalTrials.gov), the falsification criterion (BA 90% CI lower bound > 0.50), and the anti-cherry-pick rule (mechanical intersection of independent databases) were constant across all versions.

Motivation for V3.4 expansion. V3.3 ($n = 50$) was underpowered for composite significance (BA CI [0.496, 0.667] included 0.50). V3.4 added UKB-PPP instruments to rescue 75 additional genes and expanded the evaluable set to 195 pairs. The BA decreased with expansion ($0.580 \rightarrow 0.559$), confirming that additional pairs diluted rather than inflated the estimate. No disease was selected or excluded based on trial outcome distributions.

2 Mechanism Class Assignments

3 Disease-to-GWAS Mapping

For each disease, a non-UKB outcome GWAS was selected to avoid sample overlap with UKB-PPP instruments. The following table lists sources used.

Table 2: Table S2: Mechanism classification rules and examples.

Stratum	n (analyzed)	Definition and examples
Abundance-modulating	59	Drug reduces circulating protein level or blocks a soluble ligand, matching the pQTL perturbation. Examples: anti-cytokine biologics (ustekinumab, tocilizumab, anakinra), antisense oligonucleotides (mipomersen), anti-VEGF (bevacizumab), anti-complement (ravulizumab), immune checkpoint antibodies (atezolizumab).
Activity-blocking	76	Drug inhibits enzymatic activity or blocks a cell-surface receptor without necessarily changing the target protein’s measured circulating level. Examples: ACE inhibitors, Factor Xa inhibitors, thrombolytics, cholinesterase inhibitors, MMP inhibitors, PDE5 inhibitors, tyrosine kinase inhibitors.
Mixed	3	Both mechanisms apply (e.g., APP→Alzheimer’s disease, where drugs include both amyloid-clearing antibodies and secretase inhibitors).

4 Missingness Detail

5 Leave-One-Disease-Out Results

6 Loss-of-Function Burden Instrument Results

To test whether a mechanistically orthogonal genetic instrument could validate the evidence-routing dissociation, we pre-registered six analyses (V4 protocol, SHA-256 hashed before classification) using predicted loss-of-function (pLoF) burden tests from GeneBass. The pLoF burden test estimates the effect of rare coding knockouts on disease phenotypes across 450,000 UK Biobank exomes. Of 138 V3.4 pairs, 132 had usable LoF results; 6 were structurally missing due to pediatric or under-ascertained diseases (anorexia nervosa, autism spectrum disorder, juvenile idiopathic arthritis, neuroblastoma). Phenotype mapping used one canonical phenocode per disease, chosen by scientific relevance and expected case count (never by p -value), and was frozen before classification.

The fundamental constraint is statistical power. Only 7 of 132 pairs reached nominal significance ($p < 0.05$) under the LoF burden test, reflecting the rarity of pLoF carriers (~ 10 – 100 per gene in 450,000 exomes). By comparison, the pQTL instrument drew on common variants with allele frequencies of 1–50%, yielding 48 nominally significant pairs from the same 138.

Table 3: Table S3: Outcome GWAS sources by disease.

Disease	GWAS source	Overlap flag
Myocardial infarction	CARDIoGRAMplusC4D	No
Multiple sclerosis	IMSGC	No
Major depressive disorder	PGC	No
Crohn’s disease	IIBDGC	No
Ulcerative colitis	IIBDGC	No
Lung cancer	ILCCO/TRICL	No
Rheumatoid arthritis	FinnGen R10	No
Chronic kidney disease	UKB-derived	Yes
Parkinson’s disease	PGC	No
Type 2 diabetes	DIAGRAM	No
Breast cancer	BCAC	No
Prostate cancer	PRACTICAL	No

Note: Seven chronic kidney disease pairs used a UKB-derived outcome GWAS (no non-UKB alternative with adequate N) and were flagged for sensitivity analysis. Full mapping for all 24 diseases is available in the repository.

Table 4: Table S4: Missingness by mechanism stratum. The SUCCESS rate among missing abundance pairs (52.9%) exceeds the analyzed rate (30.5%), indicating non-random missingness that weakens the abundance arm. Activity-arm missingness is favorable.

Stratum	Analyzed (n)	Missing (n)	% SUCCESS (analyzed)	% SUCCESS (missing)
Abundance-modulating	59	34	30.5%	52.9%
Activity-blocking	76	22	27.6%	22.7%
Mixed	3	1	33.3%	0.0%

Table 5: Table S5: Per-disease missingness (diseases with ≥ 1 missing pair).

Disease	Analyzed	Missing	% Missing	Missing S/F
Thyroid cancer	1	15	94%	5S / 10F
JIA	1	5	83%	3S / 2F
Neuroblastoma	1	3	75%	0S / 3F
Crohn’s disease	3	4	57%	2S / 2F
Ulcerative colitis	3	3	50%	2S / 1F
Rheumatoid arthritis	7	5	42%	3S / 2F
SLE	3	2	40%	1S / 1F
Chronic kidney disease	7	3	30%	1S / 2F
Multiple sclerosis	8	3	27%	2S / 1F

Table 6: Table S6: Leave-one-disease-out BA stability. No single disease removal changes BA by more than ± 0.025 or drops BA below 0.50.

Disease dropped	n remaining	BA	Δ BA
Myocardial infarction	122	0.584	+0.024
Major depressive disorder	135	0.538	-0.021
Multiple sclerosis	130	0.546	-0.013
Chronic kidney disease	131	0.573	+0.014
Parkinson’s disease	134	0.570	+0.011
Ulcerative colitis	135	0.548	-0.011
Lung cancer	124	0.569	+0.010
JIA	137	0.549	-0.010

Table 7: Table S7: Pre-registered V4 analyses. All six returned null results. The LoF instrument is too underpowered (7/132 nominally significant) to confirm or refute the evidence-routing hypothesis.

#	Analysis	Result	p	Interpretation
1	LoF AUC (continuous)	0.538	0.49	Near chance; z-score distribution carries no signal
2	Directional concordance	51.5%	0.79	67/130 agree in sign; indistinguishable from coin flip
3	Depth-2 licensing	AUC 0.486	—	12 depth-2 pairs; 41.7% SUCCESS vs 28.8% base rate ($1.45\times$), but n too small to interpret
4	Thresholded LoF	BA 0.518	—	TP = 3, FP = 4, TN = 90, FN = 35; sensitivity ≈ 0
5	Mechanism-stratified AUC	0.586 / 0.498	0.25 / 0.99	Activity 0.586, abundance 0.498; both non-significant with overlapping CIs
6	Combined instrument	AUC 0.559	0.30	Δ AUC = +0.046 vs pQTL alone; non-significant